C++

程式語言（二）實習課

Introduction to Programming (II)

Practice Lessons

Voting System Simulation

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Platform/IDE

- Dev-C++

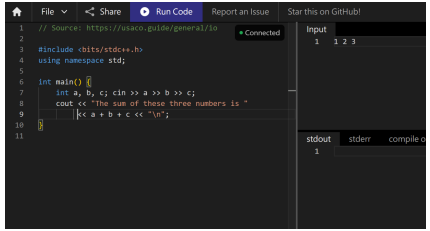
https://www.pngegg.com/en/search?q=Dev-C

- Codeblocks

https://icons8.com/icons/set/code-blocks

- OnlineGDB (https://www.onlinegdb.com/)

- Real-Time Collaborative Online IDE (https://ide.usaco.guide/)



Useful Resources

- Tutorialspoint
  - https://www.tutorialspoint.com/cplusplus/index.htm
  - Online C++ Compiler
- Programiz
  - https://www.programiz.com/cpp-programming
- LEARN C++
  - https://www.learncpp.com/
- MIT OpenCourseWare - Introduction to C++
  - https://ocw.mit.edu/courses/6-096-introduction-to-c-january-iap-2011/pages/lecture-notes/
- Learning C++ Programming
  - https://www.programiz.com/cpp-programming
- GeeksforGeeks
  - https://www.geeksforgeeks.org/c-plus-plus/

Course TAs

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Voting System Simulation (1/4)

- **Objective**:
  - Create a C++ program that simulates a voting system within a *country*. The program will involve voters casting their votes for two candidates based on the *similarity* of their *types*.
- **Description**:
  - You are required to implement a class called **Country** that manages an **array of voters**. Each voter is represented by an integer variable denoting its "**Type**", which ranges between 0 and 100. The types of voters will be input from the console.
  - The class Country also has two **static** integer variables representing two **candidates**, *A* and *B*, each with a type that will be input from the console.
  - The class should include the following functionalities:
    - **Input Voters' Types**: A member function to input the types of voters from the console.
    - **Input Candidates' Types**: A **static** member function to input the types of candidates *A* and *B* from the console.
    - **Cast Votes**: A member function where each voter calculates the difference between its type and the types of the two candidates, and votes for the candidate with the smaller difference. If the differences are equal, the voter is supposed to vote for candidate *A*.
    - **Output Winner:** A member function to output the winner based on the majority of votes received by either candidate *A* or *B*.

Voting System Simulation (2/4)

- **Requirements**:
  - Define a class Country with the following members:
    - `int voters[100]`: Array to store the types of voters (assuming a maximum of 100 voters).
    - `static int candidateAType`: Type of candidate *A*.
    - `static int candidateBType`: Type of candidate *B*.
    - `static int votesForA`: Total votes for candidate *A*.
    - `static int votesForB`: Total votes for candidate *B*.
  - Member Functions:
    - `void inputVoters(int numVoters)`: Function to input the types of voters.
    - `static void inputCandidates()`: Function to input the types of candidates *A* and *B*.
    - `void castVotes(int numVoters)`: Function to cast votes based on the types of voters and candidates.
    - `void outputWinner()`: Function to output the winner based on the votes.

Voting System Simulation (3/4)

- **Requirements (contd.)**:
  - Main Function:
    - Prompt the user to enter the number of voters.
    - Call the `inputVoters` function to input the types of voters.
    - Call the `inputCandidates` function to input the types of candidates.
    - Call the `castVotes` function to simulate the voting process.
    - Call the `outputWinner` function to display the winner.

Voting System Simulation (4/4)

```
int main() {
    Country country;
    int numVoters;
    std::cout << "number of voters: ";
    std::cin >> numVoters;

    country.inputVoters(numVoters);
    Country::inputCandidates();
    country.castVotes();
    country.outputWinner();

    return 0;
}
```

Sample input:

```
number of voters: 5
Enter types for voters:
10 20 30 40 50
Enter type for candidate A:
25
Enter type for candidate B:
35
```

Sample output:

```
Candidate A wins with 3 votes.
```

Grading Policy

- Find a TA to explain your codes. (50%)
  - The requirements will be checked.
- Submit your codes on Tronclass. (50%)